Digital Twin Concept Development as a Bridge Between Computing and Production Engineering: A Two-Course Integration in Solid Waste Management

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Abstract

This paper proposes a design methodology for cross-level, interdisciplinary course integration at the University of Brasilia (UnB), Brazil, connecting a graduate course in IT Project Management (GPTI, Applied Computing master's program) with an undergraduate Project Management course, Production Systems Project 1 (PSP1), in Production Engineering, through the shared development of Digital Twin (DT) concepts for Brasilia's Solid Waste Systems. The methodology builds on prior experiences at UnB in connecting these courses through project-based interactions, now introducing DT as a structured socio-technical coupling mechanism that simultaneously forces (i) operational grounding and (ii) disciplined digital project framing. Using the real-world context of Brasília's transition from closing an open dump site to recycling centers involving cooperatives of waste pickers, performing Selective Collection, Triage, Transformation and Commercialization, GPTI emphasizes DT as a project focus—clarifying scope, stakeholders, governance, risks, data requirements, implementation roadmap, and success criteria—while deliberately immersing computing students in operational complexity so they must design within real constraints. Conversely, PSP1 emphasizes the real system—mapping processes, constraints, and improvement opportunities across triage centers and the supply-chain concept—while requiring production engineering students to treat DTs as a technology-enabled intervention rather than an external add-on. The courses are coupled via an active, collaborative supervision model in which IT master students' mentor undergraduate teams as project leaders and requirements facilitators, and both cohorts iteratively refine shared artifacts such as system maps, requirement backlogs, risk registers, and validation criteria through joint problem-framing sessions and structured peer feedback. The proposed methodology will be applied in the next course cycle and is intended to increase authenticity, student agency, and systems thinking, yielding both a replicable course-integration approach and a set of contextualized digital twin conceptual designs grounded in the operational realities and improvement needs of Brasília's recycling centers.

Keywords: digital twin; project-based learning; active learning; engineering education; project management; solid waste management; systems thinking.

1 Introduction

Engineering education venues centered on active learning approaches, such as Problem-Based Learning (PBL) and Flipped Classroom, increasingly emphasize designs that preserve disciplinary depth while producing authentic, interdisciplinary work and assessable evidence of learning through iterative artifacts and feedback (Freeman et al., 2014; Gharajedaghi, 2011; Prince, 2004). This becomes particularly relevant for digitalization initiatives such as digital twins, which require both operational literacy (processes, constraints, stakeholder reality) and digital literacy (data, integration, governance, roadmap, validation).

This paper proposes a design methodology for a two-course integration at University of Brasília (UnB) that will be implemented in the first semester of 2026. The methodology couples (i) a graduate IT Project Management course (GPTI, Applied Computing master's program) and (ii) an undergraduate Production Engineering Project-management Course (Production Systems Project 1, PSP1). The shared challenge is the conceptual development of Digital Twin (DT) solutions for solid waste systems in Brasilia, focusing on recycling centers involving waste pickers and on improvement opportunities across the waste chain.

Importantly, this is not the first time UnB has connected these two courses. The methodological novelty here is to use DT as the bridge artifact and the shared socio-technical boundary object: a structure that simultaneously forces operational grounding (PSP1) and disciplined digital project framing (GPTI), with traceability from real needs to digital capabilities and validation criteria.

The paper contributes: (1) a replicable two-course integration architecture aligned with project approaches and PBL; and (2) an artifact-centered digital twin concept-development workflow grounded in Digital Twin Consortium (DTC) capability thinking and in PMBOK-oriented project artifact practices reflected in course materials (Digital Twin Consortium, 2020; Mello da Silva & Ishikawa, 2026; University of Brasília, 2025).

2 Related work and conceptual background

2.1 Digital twins as socio-technical systems and IT/OT/ET convergence

DTC defines DT as a virtual representation of real-world entities and processes synchronized at a specified frequency and fidelity, using real-time and historical data to represent the past and present and to simulate predicted futures (Digital Twin Consortium, 2020). In this framing, a digital twin is not simply a model or dashboard; it is a socio-technical system whose value emerges from how it supports decisions and interventions in real operations. This matters in waste management, where variability (input material quality), safety, labor constraints, logistics, and institutional governance orient what a digital system can credibly recommend, automate, or optimize.

DTC's emphasis on IT/OT/ET convergence provides a useful educational lens: Information Technology (data collection, processing, storage, communication), Operational Technology (monitoring/control of physical processes), and Engineering Technology (modeling and system design) must be integrated into a coherent intervention (University of Brasília, 2025). Pedagogically, this pushes interdisciplinary collaboration: computing students cannot treat operations as a "domain black box," and production engineering students cannot treat digitalization as an external layer that appears after process mapping.

2.2 IT/OT/ET convergence and why it is central to "Digital Twin-first"

DTs become most challenging and most valuable precisely when they sit at the intersection of IT (Information Technology systems and data/analytics), OT (Operational Technology such as sensors, automation, and industrial control), and ET (Engineering Technology such as models, design intent, and system engineering artifacts) (Digital Twin Consortium, 2020). In practice, DT must reconcile heterogeneous data sources and time scales: IT often operates on business rhythms and transactional correctness; OT operates on real-time constraints, safety, and reliability; ET operates on design fidelity, physics, and lifecycle reasoning. When these worlds converge, requirements also converge: integration and data services cannot be separated from trustworthiness, and intelligence cannot be separated from governance and operational safety (Brazil, 2010; Gharajedaghi, 2011; Ministério do Meio Ambiente e Mudança do Clima, 2024).

The instructional design leverages this convergence to make students practice realistic reasoning. Real-life System-in-Question (RSQ) and Digital System-in-Question (DSQ) boundaries must explicitly include where IT ends and OT begins (e.g., what data is available, at what cadence, under what security constraints) and how ET representations are maintained (e.g., whether the DSQ includes engineering models, assumptions, and their versioning).

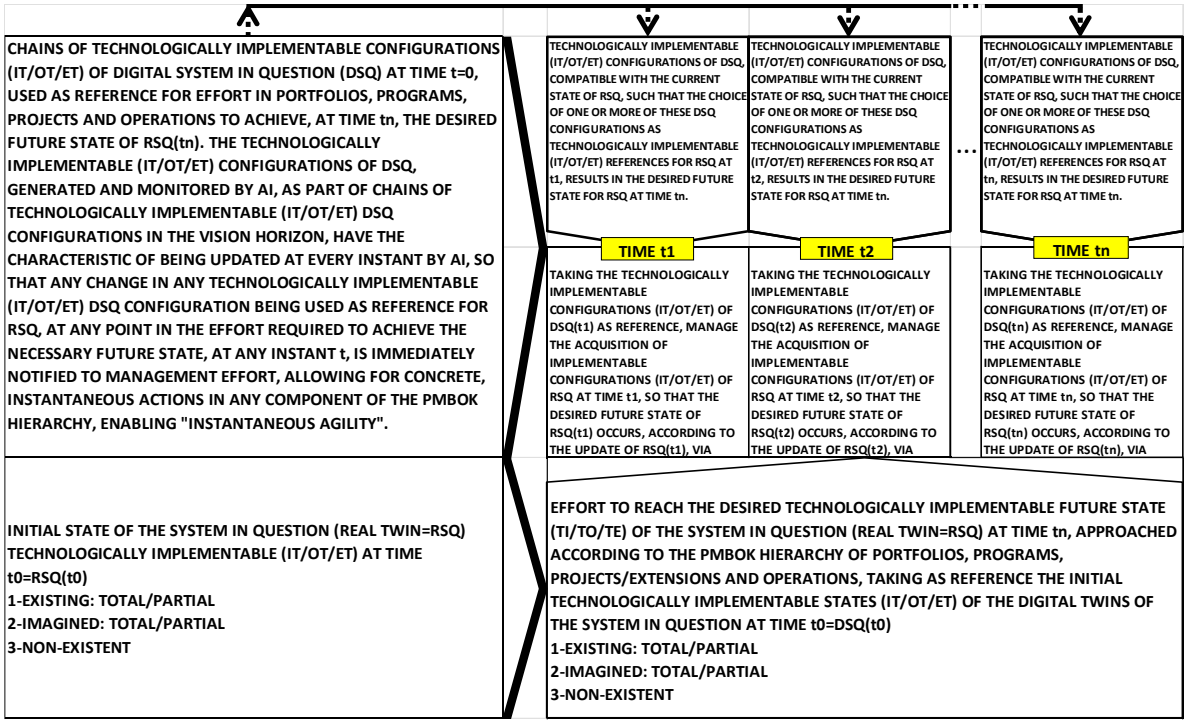


Figure 1. Digital Twin as Reference for Real Twin

As shown in Figure 1, DT is the central coupling mechanism and shared boundary object: it requires operational grounding and capability-oriented digital specification in the same artifact package. A DT is most strategic when used not just as a copy of the present, but as a reference for defending decisions and guiding change toward defined future states, obtained from AI's role in building nexus for scenario creation.

2.3 Capability-oriented specification

Student teams often propose a "platform" without clarifying required capabilities and dependencies. To mitigate this, the methodology applies a capability-oriented specification approach grounded in the DTC Capabilities Periodic Table (DTC/CPT), referenced in the GPTI plan (Mello da Silva & Ishikawa, 2026). The DTC/CPT structures DT thinking into capability groups (DS-Data Services, IR-Integration, IC-Intelligence, UX-User Experience, MG-Management, and TW-Trustworthiness). When teams incorporate AI agents to support monitoring, reporting, or decision recommendations, the capability thinking is extended with the DTC AI Agent Capabilities Periodic Table (AIA/CPT) (Mello da Silva & Ishikawa, 2025, 2026).

2.4 Active learning, and cross-level mentoring

The educational design aligns with evidence on PBL and Flipped Classroom active learning approaches: structured problem solving, feedback, and iterative refinement generally outperform passive approaches in STEM contexts (Freeman et al., 2014; Gharajedaghi, 2011; Prince, 2004). Cross-level coupling also aligns with peer tutoring/near-peer learning literature, where near-peers can scaffold learning through guided questioning and feedback when roles and safeguards preserve agency (Hmelo-Silver, 2004; Topping, 1996).

3 Educational context and design goals

3.1 Courses and institutional context

The collaboration occurs at UnB, within two complementary programs: Applied Computing (graduate) and Production Engineering (undergraduate). The graduate course (GPTI) targets IT project management with explicit emphasis on teamwork, a systems approach, and PBL/Flipped classroom methods (Mello da Silva & Ishikawa, 2026), focusing on the DT while the undergraduate course (PSP1) targets the RT. Both courses are also directed to project management, emphasizing operational understanding, constraints, and improvement opportunities.

3.2 Domain context: waste management and recycling centers

The RT is based on the RSQ, which in the current version of the courses is the recycling centers of Brasilia's solid waste system, mainly operated by cooperatives of waste pickers, with a focus on Selective Collection, Triage, Value-adding Transformation opportunities, and Commercialization. Brazil's National Solid Waste Policy (Brazil, 2010) and the corresponding regulation (Ministério do Meio Ambiente e Mudança do Clima, 2024) are the ground of public relevance for this subject.

In the Brasília solid waste context adopted as the Real System in Question (RSQ/Real-life Twin), the student work is grounded in the Law No. 12,305/2010, which establishes Brazil's National Solid Waste Policy, complemented by the regulatory framework of GM/MMA Ordinance No. 1,250 (December 13, 2024) (Brazil, 2010; Ministério do Meio Ambiente e Mudança do Clima, 2024). This ordinance establishes the Incentive mechanism for the Recycling Industry and Production Chain. In line with Article 3, the RSQ is framed as a set of stages that extend beyond basic sorting/beneficiation, explicitly including processing and manufacturing as pathways for the industrial transformation of recyclable materials into higher value-added products. This RSQ framing connects directly to the ordinance's targets in Article 4—capacity building, training, and technical advisory (including national and international exchanges); incubation of cooperatives and solidarity-based social enterprises in recycling; implementation and adaptation of physical infrastructure for cooperatives and waste-picker associations; acquisition of equipment for selective collection, beneficiation, treatment, and recycling; organization and support for marketing networks and production chains; strengthening waste pickers' participation in recycling value chains; and technology development to add value to collection and recycling work—and to Article 6 by treating waste-picker enterprises and/or autonomous waste pickers as a structuring element of the execution scope. In this way, the RSQ/Real Twin is characterized not only by material flows and operational constraints, but also by mandated forms of participation, capacity building, and value-adding transformation, which in turn define the decision boundaries, governance needs, and evidence requirements that the corresponding Digital System in Question (DSQ/Digital Twin) must respect when representing and supporting the real system.

The Incentive mechanism for the Recycling Industry and Production Chain commission has received 952 project proposals in 2025, totaling 2,2 billion Brazilian reais in proposals aiming for investments throughout 27 Brazilian federal units (Ministério do Meio Ambiente e Mudança do Clima, 2025).

4 Two-course integration architecture

4.1 Role structure and coupling mechanism

The methodology uses an explicit role structure. As already stated, GPTI teams focus on project aspects of

DSQs and DTs, while PSP1 teams have RSQs and RTs as targets.

The cross-course interaction is organized around a shared paired description of the RSQ as a RT and its corresponding DSQ as a DT, with clearly defined reciprocal roles between cohorts. At the first class of PSP1, graduate GPTI students send representatives to introduce to the undergraduate students core project-management concepts, encompassing foundational DT framing, aligned with Digital Twin Consortium concepts, and a practical method for producing a tailored project guide for DTs of solid waste systems components, grounded in PMBOK editions (4th, 6th, 7th, and 8th) and the PMI Guide to Business Analysis (PMI, 2017b), expressed in an adapted PMBOK-extension format (PMI, 2008, 2017a, 2021, 2025). To anchor this transfer in concrete evidence rather than abstraction, GPTI representatives also present selected results and lessons learned from the previous GPTI offering, accessed through the instructor, as worked examples of how digital-twin project reasoning can be packaged and defended.

In the opposite direction, PSP1 sends representatives to the first class of GPTI to present the functional groups corresponding to the four components (Selective Collection, Triage, Transformation and Commercialization) of RSQ, as both a production system and a human activity system, emphasizing real operational flows, constraints, coordination practices, and feasibility boundaries that the DSQ must respect, as in Figure 2.

In addition, PSP1 students share details relative to the legal and regulatory aspects regarding the RSQ, explicitly framing Brazil's National Solid Waste Policy (Law No. 12,305/2010) and the GM/MMA Ordinance No. 1,250 of 13 December 2024.

A key message communicated by GPTI to PSP1 is that project management is not "only PMBOK": the integration explicitly treats PMBOK guidance as a basis for tailoring and extends PMBOK8 to place the digital twin as a central element, while foregrounding SIPA (S-Sustainability, I-Innovation, P-Privacy, and A-Business Analysis) as essential lenses for socio-technical digital-twin projects in solid waste contexts.

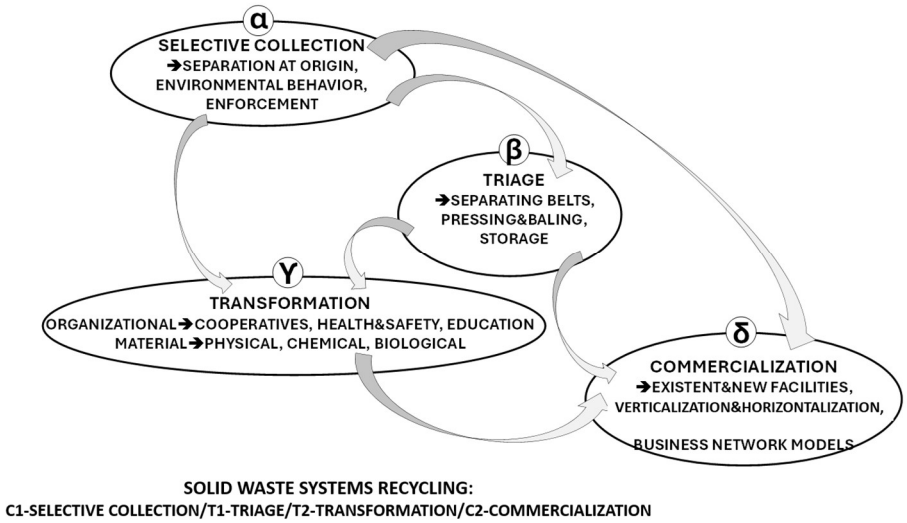


Figure 2. Solid Waste System Functional groups CTTC

4.2 Deliverables as courses coupling points

Figure 3 operationalizes the coupling between GPTI and PSP1 through a shared deliverable chain that functions as the interface between cohorts. On the GPTI side, the flow starts from PMBOK8 and evolves into an explicit extension structure (PMBOK8 to GPH.SIPA.S), adding SIPA dimensions—S-Sustainability, I-Innovation, P-Privacy, and A-Business Analysis—so that tailoring becomes an explicit methodological step rather than an

informal adaptation. This extension is then advanced through group work (GPH.SIPA.S), moving from partial outputs to a consolidated version and bridging toward PMBOK8 framing, while remaining anchored in operational stakeholder perspectives (e.g., product developers, waste-picker cooperatives, and landfill operators). In parallel, the RSQ is framed as both a production system and a human activity system by locating solid waste within the broader sanitation system and structuring the solid-waste domain through P3RD priorities (Prevention, Reduction, Reuse, Recycling, Disposal), complemented by the "Digital Mobilizer" role that connects operational reality to digital project framing.

The coupling occurs at specific checkpoints where deliverables are intentionally designed as shared reference points rather than shared submissions. TI01 consolidates early deliverables (P2A/P2B) that stabilize a common RSQ description and stakeholder grounding, enabling PSP1 representatives to brief GPTI on functional groups and constraints, and GPTI representatives to brief PSP1 on project management and digital-twin framing. TI02A structures mid-semester deliverables (P3A/P3B/P3C/P3D) that translate the RSQ description into the extended PMBOK logic (GPH.SIPA.S), culminating in a tailored project-management guide for solid-waste systems and operational stakeholders. TI02B marks the late-stage coupling through P4A (delivery of the tailored guide) and P4B (initial milestones and pre-project elements), including needs and gaps, business case, benefits management plan, project charter, and a baseline project management plan. Finally, TE01 (P6C1/P6C2) is the closing joint tele-event, where leaders from both courses present a structured synthesis of convergences between RSQ and DSQ perspectives—mirroring the classes of both courses as a single coordinated team while each course still completes its own independent final work.

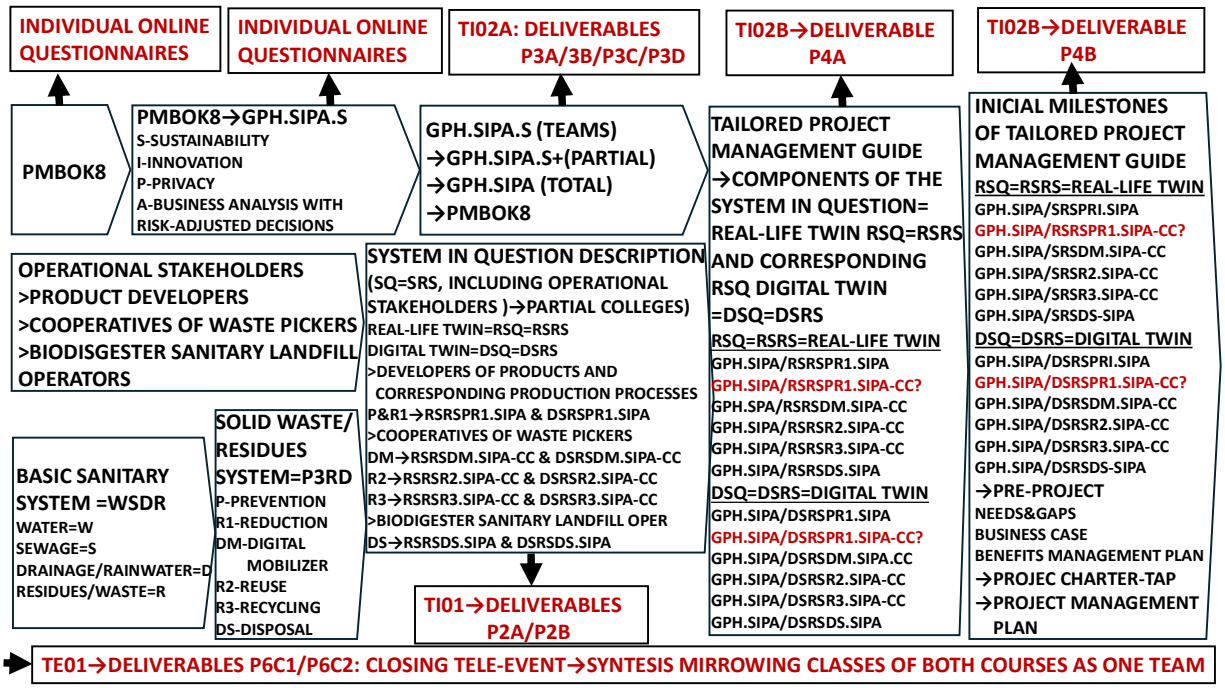


Figure 3. Deliverables

Although each course maintains its own internal assignments and assessment, the integration is sustained through a small set of structured joint coordination touchpoints, involving team-leaders and college-leaders presentations, under the guidance of teaching assistants (tutors). Due to different course schedules, the two confirmed synchronized moments are the beginning and the end: PSP1 runs for 16 weeks and GPTI for 8 weeks, so GPTI starts mid-way through the PSP1 semester, which makes it fully feasible to align GPTI's kickoff with PSP1's ongoing work and to ensure both courses conclude together. At the beginning of PSP1, the initial domain and methodological context is introduced by representatives connected to GPTI (leveraging results

and learning artifacts from the immediate previous GPTI offering, provided via the instructor), setting a concrete baseline for how RSQ/DSQ reasoning and tailored project-management guidance will be treated.

To ensure operational grounding, capability thinking is complemented with a shared description pattern also present in course materials: Function, Structure, Process, and Context (F/S/P/C) (Gharajedaghi, 2011; University of Brasília, 2025). F/S/P/C provides a stable anchor for systems thinking: teams define boundaries, interfaces, constraints, and decision points before deriving digital requirements. As shown in Figure 4, the process of interconnection between the GPTI and PSP1 courses becomes operational mainly during task TI01, encompassing deliverables P2A and P2B.

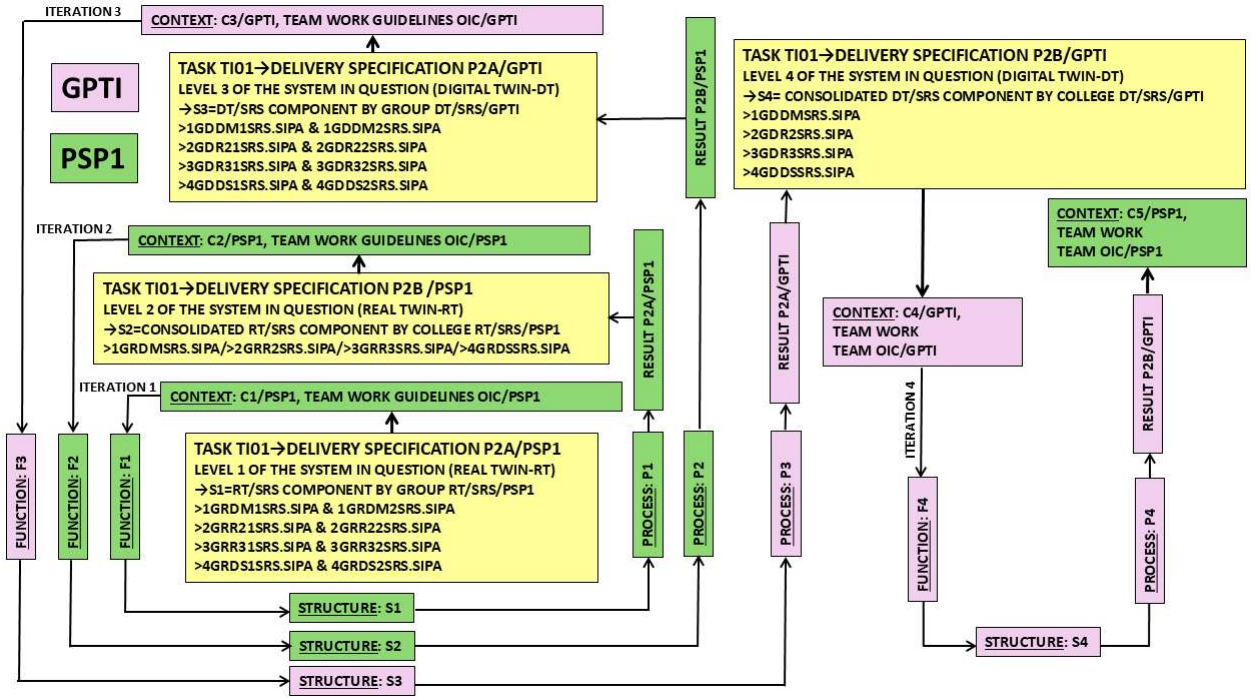


Figure 4. The two courses link – PSP1 and GPTI

During the middle of the semester, coordination is maintained through monitor-led meetings and, when needed, additional meetings among student team leaders to prepare and refine the four joint presentation themes—(i) RSQ/DSQ descriptions, (ii) PMBOK foundations and tailoring logic with the inclusion of SIPA and DTC concepts, (iii) applying the tailored adapted PMBOK extension format to the corresponding RSQ/DSQ descriptions, and (iv) the closing joint tele-event. The tele-event serves as the final cross-course presentation, where PSP1 and GPTI teams explicitly surface convergences between operational system understanding and digital-twin project framing, while preserving the independence of each course's deliverables.

5 Conclusion

The two-course integration uses the Digital Twin not as an "extra topic," but as a disciplined bridge that forces mutual intelligibility between Applied Computing and Production Engineering. By pairing the Real System in Question (RSQ/Real-life Twin) with its Digital System in Question (DSQ/Digital Twin) and by structuring interaction through reciprocal representative sessions and a shared deliverable chain, the design makes cross-level collaboration auditable: PSP1 contributes operational descriptions grounded in production-system and human-activity realities (RSQ), while GPTI contributes with the Digital Twin Consortium framing. Both courses operate based on project-tailoring logic anchored in PMBOK editions and extended through the inclusion of

SIPA (Sustainability, Innovation, Privacy, and Business Analysis). The result is an integration that preserves disciplinary depth while creating a common language for socio-technical design decisions, explicitly tied to regulatory constraints and stakeholder participation in Brasília's solid waste recycling context.

From an engineering-education perspective, the core contribution is a replicable coupling mechanism where learning is made visible through structured checkpoints and deliverables rather than informal collaboration. The cadence of joint presentations (system description, PMBOK foundations, applying the tailored guide to the RSQ/DSQ pair, and the closing tele-event synthesis) provides a practical pathway for aligning PMBOK8 project approach and active learning principles with credible evidence of student reasoning and integration quality. More broadly, the approach offers a transferable pattern for other domains in which digitalization must be grounded in real operational constraints: when the coupling is anchored in a shared RSQ/DSQ description and a tailored project-management guide, students practice not only "designing a technology," but also packaging, defending, and governing a socio-technical pathway that can realistically be executed.

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